

I Additional examples for repetitive failures by LLMs

To supplement the qualitative examples in Figure 3 (Section 4), we provide models’ responses that consistently produce erroneous outputs for certain prompts in Figures 19 and 24. The examples illustrate that Qwen2.5-Math-7B-Instruct exhibits repetitive errors for certain ratios of sugar teaspoons to water cups, frequently invoking a spurious “standard US measurement” heuristic to convert cups into tablespoons or teaspoons. For gpt-oss-20b-low on Template 7, the model shows striking failures on the basic arithmetic $84 \times (2/7)$, while being generally robust to other fraction values. We additionally provide dominant failure cases for all model-template pairs and observe various error types. These include simple arithmetic errors in addition/subtraction (18, 26), multiplication/division (16, 20, 17, 23, 25, 27), transcription (21), unnecessary approximation (22), and problem misinterpretation (24). Failure-inducing parameter values and the onset of erroneous reasoning in the generation traces are highlighted in red.

Prompt (Template 0)

A {dolphin} is {1276} feet long. {11} {remoras} attach to it, each measuring {348} inches in length. What percentage of the {dolphin}'s body length is the combined length of the {remoras}?

First Output (Qwen2.5-Math-7B-Instruct) (Incorrect)

To determine what percentage of the dolphin's body length is the combined length of the remoras, we need to follow these steps: ... Now, we perform the division:

$$\frac{319}{1276} \approx 0.2475$$

Then, we multiply by 100 to convert the decimal to a percentage:

$$0.2475 \times 100 = 24.75$$

Therefore, the combined length of the remoras is 24.75 percent of the dolphin's body length.

Second Output (Correct)

To determine what percentage of the dolphin's body length is the combined length of the remoras, we need to follow these steps: ... Perform the division and multiplication:

$$\frac{319}{1276} \approx 0.25$$

$$0.25 \times 100 = 25$$

Therefore, the combined length of the remoras is 25 percent of the dolphin's body length.

Third Output (Correct)

To determine what percentage of the dolphin's body length is the combined length of the remoras, we need to follow these steps: ...

$$\left(\frac{319}{1276} \right) \times 100 \approx 25 \text{ percent}$$

Thus, the combined length of the remoras is 25 percent of the dolphin's body length.

Fourth Output (Incorrect)

To determine what percentage of the dolphin's body length is the combined length of the remoras, we need to follow these steps: ... Now, we perform the division and multiplication:

$$\frac{319}{1276} \approx 0.247$$

$$0.247 \times 100 = 24.7$$

Therefore, the combined length of the remoras is approximately 24.7% of the dolphin's body length. The final answer is:

$$\boxed{24.7}$$

Fifth Output (Incorrect)

To determine what percentage of the dolphin's body length is the combined length of the remoras, we need to follow these steps: ...

$$\left(\frac{319 \text{ feet}}{1276 \text{ feet}} \right) \times 100 = \frac{31900}{1276} \approx 24.73\%$$

Therefore, the combined length of the remoras is 24.73 % of the dolphin's body length.

Sixth Output (Correct)

To determine what percentage of the dolphin's body length is the combined length of the remoras, we need to follow these steps: ...

$$\text{Percentage} = \left(\frac{319 \text{ feet}}{1276 \text{ feet}} \right) \times 100 \approx 25\%$$

Therefore, the combined length of the remoras is 25 % of the dolphin's body length.

Seventh Output (Incorrect)

To determine what percentage of the dolphin's body length is the combined length of the remoras, we need to follow these steps: ...

$$\text{Percentage} = \left(\frac{319 \text{ feet}}{1276 \text{ feet}} \right) \times 100 \approx 24.72\%$$

Therefore, the combined length of the remoras is 24.72 % of the dolphin's body length.

Eighth Output (Incorrect)

To determine what percentage of the dolphin's body length is the combined length of the remoras, we need to follow these steps: ... Perform the division and multiplication:

$$\frac{319}{1276} \approx 0.247$$

$$0.247 \times 100 = 24.7$$

Therefore, the combined length of the remoras is approximately 24.7 % of the dolphin's body length.

Answer:25

Figure 16: Eight independently generated outputs from Qwen2.5-Math-7B-Instruct evaluated on Template 0. Error type: wrong division.

Prompt (Template 1) A {coastal mist} rolls in from the ocean to cover {Harborview}. It takes {33} minutes to cover every {4} miles of the city. If the city is {28} miles across from the oceanfront to the opposite inland edge, how many minutes will it take for the {coastal mist} to cover the whole city?
First Output (Qwen2.5-Math-7B-Instruct) (Correct) To determine how many minutes it will take for the marine layer to cover the entire city, we start by identifying the rate at which the marine layer rolls in. The marine layer covers 4 miles in 33 minutes. We need to find out how many minutes it will take to cover 28 miles. ... $8.25 \times 28 = 8.25 \times (30 - 2) = 8.25 \times 30 - 8.25 \times 2$ $8.25 \times 30 = 247.5$ $8.25 \times 2 = 16.5$ $247.5 - 16.5 = 231$ Therefore, it will take the marine layer 231 minutes to cover the whole city.
Second Output (Incorrect) To determine how many minutes it will take for the marine layer to cover the entire city, we start by identifying the rate at which the marine layer rolls in. The marine layer covers 4 miles in 33 minutes. We need to find out how many minutes it will take to cover 28 miles. ... $\text{Time to cover 28 miles} = 8.25 \text{ minutes per mile} \times 28 \text{ miles} = 229.5 \text{ minutes}$ Therefore, it will take the marine layer 229.5 minutes to cover the whole city.
Third Output (Correct) To determine how many minutes it will take for the marine layer to cover the entire city, we need to follow these steps: ... Since it takes 33 minutes to cover each 4-mile segment, we multiply the number of segments by the time per segment: $\text{Total time} = 7 \text{ segments} \times 33 \text{ minutes/segment} = 231 \text{ minutes}$ Therefore, it will take 231 minutes for the marine layer to cover the whole city.
Fourth Output (Incorrect) To determine how many minutes it will take for the marine layer to cover the entire city, we start by identifying the rate at which the marine layer rolls in. The marine layer covers 4 miles in 33 minutes. We need to find out how long it will take to cover 28 miles. ... $8.25 \text{ minutes per mile} \times 28 \text{ miles} = 229.5 \text{ minutes}$ Therefore, it will take the marine layer 229.5 minutes to cover the whole city.
Fifth Output (Incorrect) To determine how many minutes it will take for the marine layer to cover the entire city, we start by finding the rate at which the marine layer rolls in. The marine layer covers 4 miles in 33 minutes. We need to find out how many minutes it takes to cover 28 miles. ... $8.25 \text{ minutes per mile} \times 28 \text{ miles} = 229.5 \text{ minutes}$ Therefore, it will take the marine layer 229.5 minutes to cover the whole city.
Sixth Output (Incorrect) To determine how many minutes it will take for the marine layer to cover the entire city, we start by identifying the rate at which the marine layer rolls in. The marine layer covers 4 miles in 33 minutes. We need to find out how many minutes it will take to cover 28 miles. ... $8.25 \text{ minutes per mile} \times 28 \text{ miles} = 229.5 \text{ minutes}$ Therefore, it will take the marine layer 229.5 minutes to cover the whole city.
Seventh Output (Correct) To determine how many minutes it will take for the marine layer to cover the entire city, we start by finding the rate at which the marine layer covers the city. The problem states that it takes 33 minutes to cover every 4 miles. ... Since it takes 33 minutes to cover each 4-mile segment, we multiply the number of segments by the time it takes to cover each segment: $\text{Total time} = 7 \text{ segments} \times 33 \text{ minutes/segment} = 231 \text{ minutes}$ Thus, the total time it will take for the marine layer to cover the whole city is: $\boxed{231}$
Eighth Output (Incorrect) To determine how many minutes it will take for the marine layer to cover the entire city, we start by identifying the rate at which the marine layer rolls in. The marine layer covers 4 miles in 33 minutes. We need to find out how many minutes it will take to cover 28 miles. ... $8.25 \text{ minutes per mile} \times 28 \text{ miles} = 229.5 \text{ minutes}$ Therefore, it will take the marine layer 229.5 minutes to cover the whole city. Answer:231

Figure 17: Eight independently generated outputs from Qwen2.5-Math-7B-Instruct evaluated on Template 1. Error type: wrong multiplication.